LAPORAN PRAKTIKUM DEVSECOPS
KELOMPOK 3

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PROGRAM STUDI D4 TEKNIK INFORMATIKA

POLITEKNIK ELEKTRONIKA NEGERI SURABAYA (PENS)

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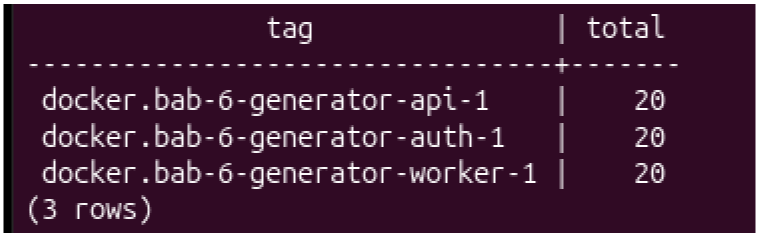
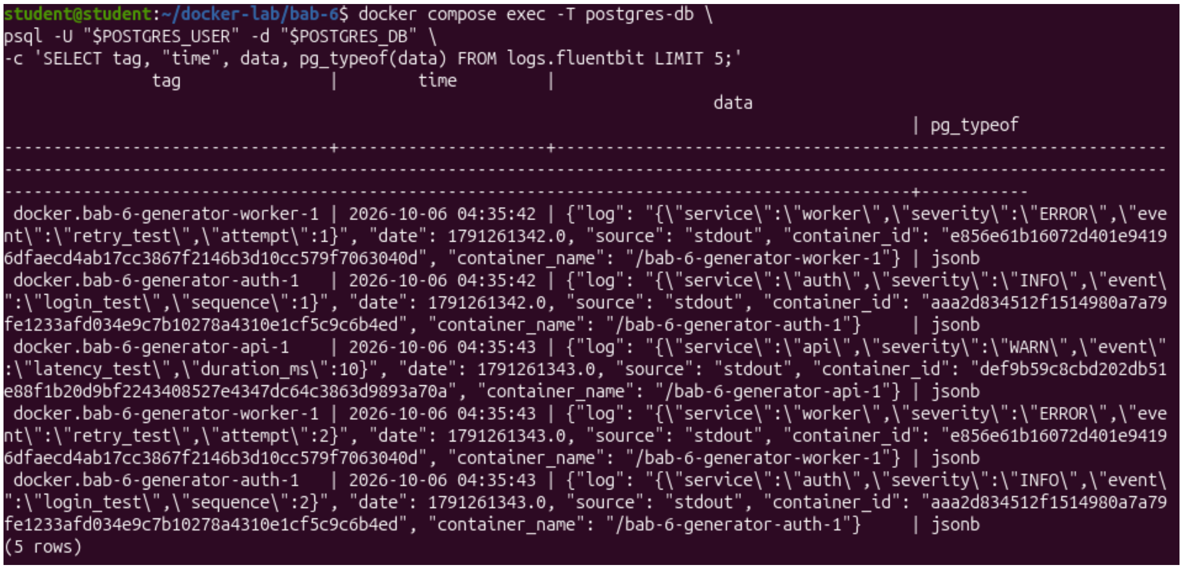
Verifikasi dan Skenario Pengujian

[ ] Fluent Bit menerima log dari minimal tiga container.



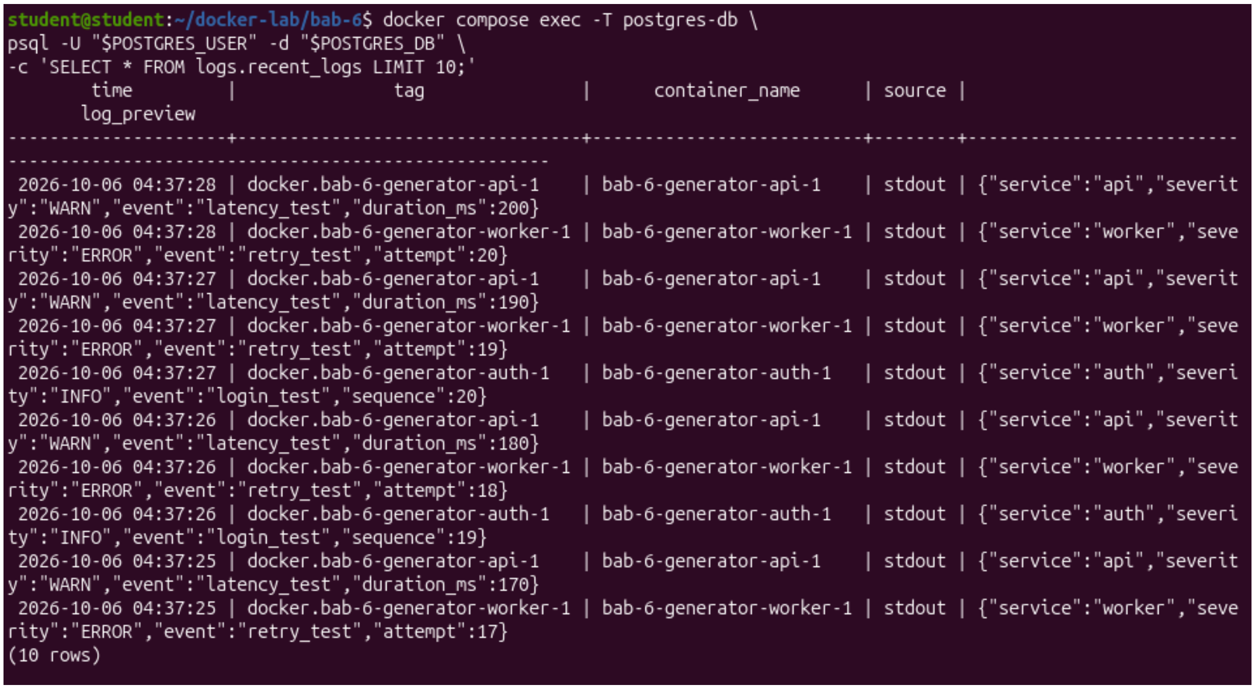
```
               tag                | total
----------------------------------+-------
 docker.bab-6-generator-api-1     |    20
 docker.bab-6-generator-auth-1    |    20
 docker.bab-6-generator-worker-1  |    20
(3 rows)
```

[ ] Tabel logs.fluentbit berisi data dengan tag, time, dan data JSONB.



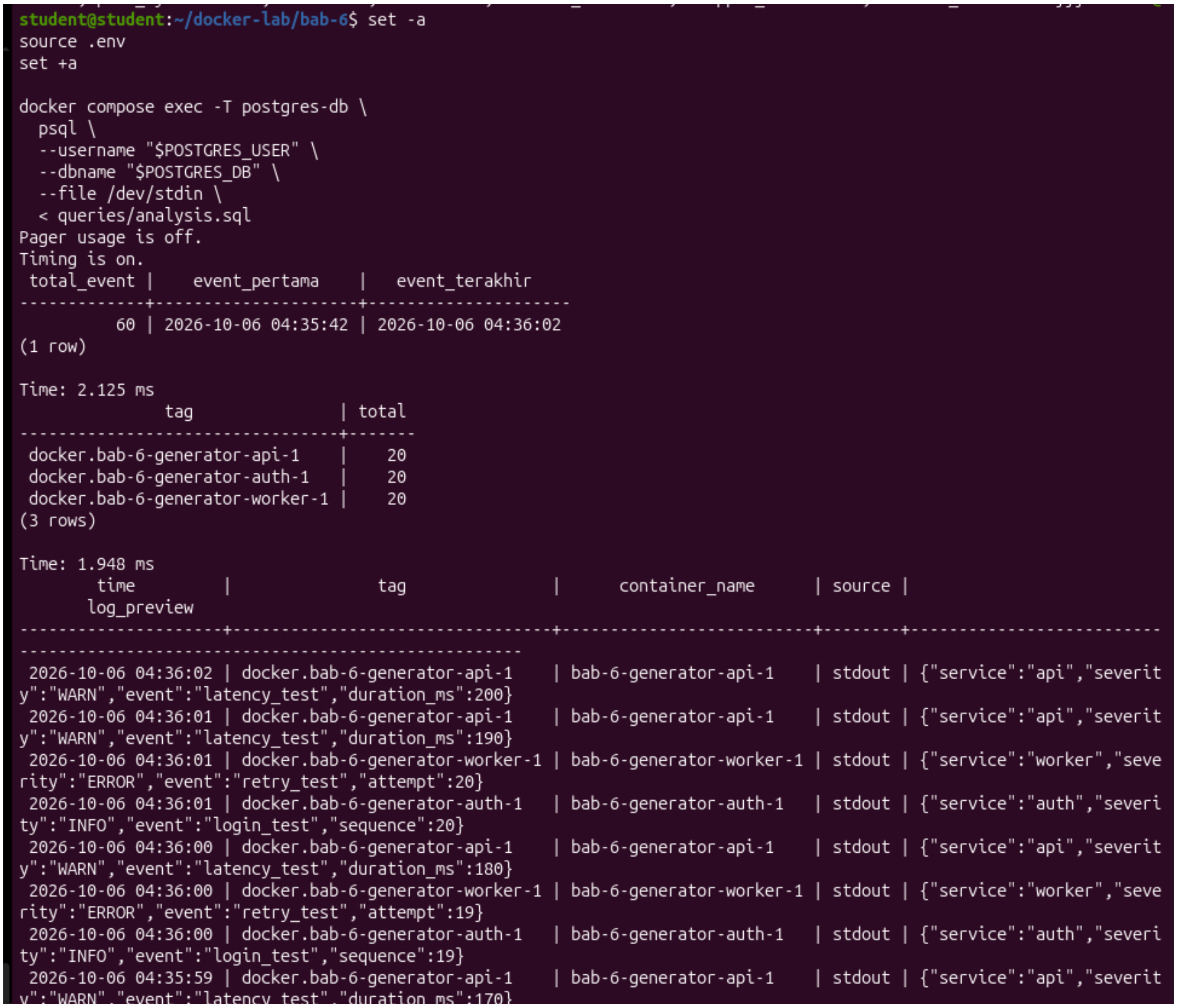
```
student@student:~/docker-lab/bab-6$ docker compose exec -T postgres-db \
psql -U "$POSTGRES_USER" -d "$POSTGRES_DB" \
-c 'SELECT tag, "time", data, pg_typeof(data) FROM logs.fluentbit LIMIT 5;'
               tag               |        time         |
                                                               data
                                                                                              | pg_typeof
---------------------------------+---------------------+---------------------------------------------------------------
----------------------------------------------------------------------------------------------------------------------
----------------------------------------------------------------------------------------------+-----------
 docker.bab-6-generator-worker-1 | 2026-10-06 04:35:42 | {"log": "{\"service\":\"worker\",\"severity\":\"ERROR\",\"eve
nt\":\"retry_test\",\"attempt\":1}", "date": 1791261342.0, "source": "stdout", "container_id": "e856e61b16072d401e9419
6dfaecd4ab17cc3867f2146b3d10cc579f7063040d", "container_name": "/bab-6-generator-worker-1"} | jsonb
 docker.bab-6-generator-auth-1   | 2026-10-06 04:35:42 | {"log": "{\"service\":\"auth\",\"severity\":\"INFO\",\"event\
":\"login_test\",\"sequence\":1}", "date": 1791261342.0, "source": "stdout", "container_id": "aaa2d834512f1514980a7a79
fe1233afd034e9c7b10278a4310e1cf5c9c6b4ed", "container_name": "/bab-6-generator-auth-1"}     | jsonb
 docker.bab-6-generator-api-1    | 2026-10-06 04:35:43 | {"log": "{\"service\":\"api\",\"severity\":\"WARN\",\"event\"
:\"latency_test\",\"duration_ms\":10}", "date": 1791261343.0, "source": "stdout", "container_id": "def9b59c8cbd202db51
e88f1b20d9bf2243408527e4347dc64c3863d9893a70a", "container_name": "/bab-6-generator-api-1"} | jsonb
 docker.bab-6-generator-worker-1 | 2026-10-06 04:35:43 | {"log": "{\"service\":\"worker\",\"severity\":\"ERROR\",\"eve
nt\":\"retry_test\",\"attempt\":2}", "date": 1791261343.0, "source": "stdout", "container_id": "e856e61b16072d401e9419
6dfaecd4ab17cc3867f2146b3d10cc579f7063040d", "container_name": "/bab-6-generator-worker-1"} | jsonb
 docker.bab-6-generator-auth-1   | 2026-10-06 04:35:43 | {"log": "{\"service\":\"auth\",\"severity\":\"INFO\",\"event\
":\"login_test\",\"sequence\":2}", "date": 1791261343.0, "source": "stdout", "container_id": "aaa2d834512f1514980a7a79
fe1233afd034e9c7b10278a4310e1cf5c9c6b4ed", "container_name": "/bab-6-generator-auth-1"}     | jsonb
(5 rows)
```

[ ] View recent_logs menampilkan log terbaru.



```
student@student:~/docker-lab/bab-6$ docker compose exec -T postgres-db \
psql -U "$POSTGRES_USER" -d "$POSTGRES_DB" \
-c 'SELECT * FROM logs.recent_logs LIMIT 10;'
        time         |               tag               |      container_name      | source |
        log_preview
---------------------+---------------------------------+--------------------------+--------+-------------------------
-----------------------------------------------------
 2026-10-06 04:37:28 | docker.bab-6-generator-api-1    | bab-6-generator-api-1    | stdout | {"service":"api","severit
y":"WARN","event":"latency_test","duration_ms":200}
 2026-10-06 04:37:28 | docker.bab-6-generator-worker-1 | bab-6-generator-worker-1 | stdout | {"service":"worker","seve
rity":"ERROR","event":"retry_test","attempt":20}
 2026-10-06 04:37:27 | docker.bab-6-generator-api-1    | bab-6-generator-api-1    | stdout | {"service":"api","severit
y":"WARN","event":"latency_test","duration_ms":190}
 2026-10-06 04:37:27 | docker.bab-6-generator-worker-1 | bab-6-generator-worker-1 | stdout | {"service":"worker","seve
rity":"ERROR","event":"retry_test","attempt":19}
 2026-10-06 04:37:27 | docker.bab-6-generator-auth-1   | bab-6-generator-auth-1   | stdout | {"service":"auth","severi
ty":"INFO","event":"login_test","sequence":20}
 2026-10-06 04:37:26 | docker.bab-6-generator-api-1    | bab-6-generator-api-1    | stdout | {"service":"api","severit
y":"WARN","event":"latency_test","duration_ms":180}
 2026-10-06 04:37:26 | docker.bab-6-generator-worker-1 | bab-6-generator-worker-1 | stdout | {"service":"worker","seve
rity":"ERROR","event":"retry_test","attempt":18}
 2026-10-06 04:37:26 | docker.bab-6-generator-auth-1   | bab-6-generator-auth-1   | stdout | {"service":"auth","severi
ty":"INFO","event":"login_test","sequence":19}
 2026-10-06 04:37:25 | docker.bab-6-generator-api-1    | bab-6-generator-api-1    | stdout | {"service":"api","severit
y":"WARN","event":"latency_test","duration_ms":170}
 2026-10-06 04:37:25 | docker.bab-6-generator-worker-1 | bab-6-generator-worker-1 | stdout | {"service":"worker","seve
rity":"ERROR","event":"retry_test","attempt":17}
(10 rows)
```

[ ] Query distribusi per tag dan pencarian keyword berjalan.



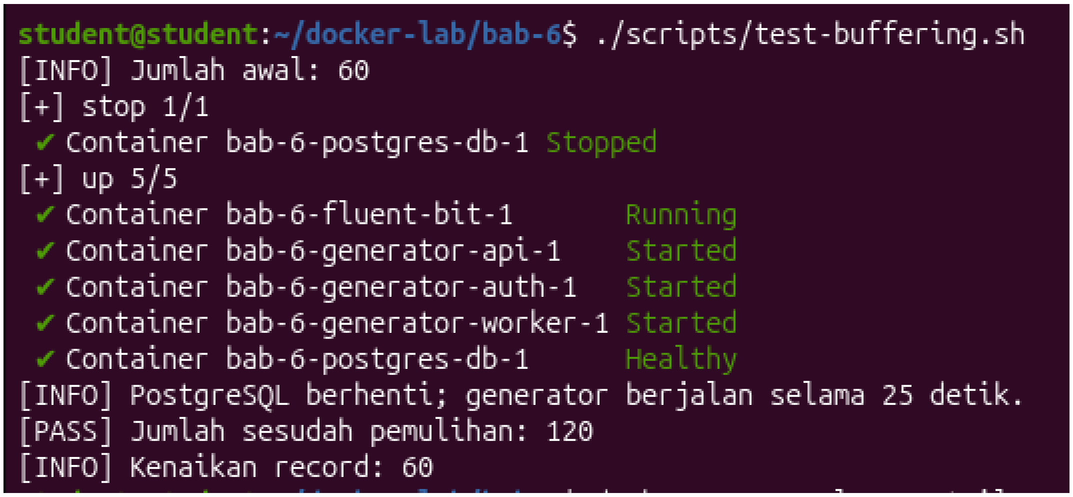
```
student@student:~/docker-lab/bab-6$ set -a
source .env
set +a

docker compose exec -T postgres-db \
  psql \
  --username "$POSTGRES_USER" \
  --dbname "$POSTGRES_DB" \
  --file /dev/stdin \
  < queries/analysis.sql
Pager usage is off.
Timing is on.
 total_event |    event_pertama    |    event_terakhir
-------------+---------------------+---------------------
          60 | 2026-10-06 04:35:42 | 2026-10-06 04:36:02
(1 row)

Time: 2.125 ms
              tag               | total
--------------------------------+-------
 docker.bab-6-generator-api-1    |    20
 docker.bab-6-generator-auth-1   |    20
 docker.bab-6-generator-worker-1 |    20
(3 rows)

Time: 1.948 ms
        time         |               tag               |      container_name      | source |
       log_preview
---------------------+---------------------------------+--------------------------+--------+-------------------------
----------------------------------------------------
 2026-10-06 04:36:02 | docker.bab-6-generator-api-1    | bab-6-generator-api-1    | stdout | {"service":"api","severit
y":"WARN","event":"latency_test","duration_ms":200}
 2026-10-06 04:36:01 | docker.bab-6-generator-api-1    | bab-6-generator-api-1    | stdout | {"service":"api","severit
y":"WARN","event":"latency_test","duration_ms":190}
 2026-10-06 04:36:01 | docker.bab-6-generator-worker-1 | bab-6-generator-worker-1 | stdout | {"service":"worker","seve
rity":"ERROR","event":"retry_test","attempt":20}
 2026-10-06 04:36:01 | docker.bab-6-generator-auth-1   | bab-6-generator-auth-1   | stdout | {"service":"auth","severi
ty":"INFO","event":"login_test","sequence":20}
 2026-10-06 04:36:00 | docker.bab-6-generator-api-1    | bab-6-generator-api-1    | stdout | {"service":"api","severit
y":"WARN","event":"latency_test","duration_ms":180}
 2026-10-06 04:36:00 | docker.bab-6-generator-worker-1 | bab-6-generator-worker-1 | stdout | {"service":"worker","seve
rity":"ERROR","event":"retry_test","attempt":19}
 2026-10-06 04:36:00 | docker.bab-6-generator-auth-1   | bab-6-generator-auth-1   | stdout | {"service":"auth","severi
ty":"INFO","event":"login_test","sequence":19}
 2026-10-06 04:35:59 | docker.bab-6-generator-api-1    | bab-6-generator-api-1    | stdout | {"service":"api","severit
y":"WARN","event":"latency_test","duration_ms":170}
```

[ ] Mahasiswa memahami apa yang terjadi jika collector mati sementara.



```
student@student:~/docker-lab/bab-6$ ./scripts/test-buffering.sh
[INFO] Jumlah awal: 60
[+] stop 1/1
 ✔ Container bab-6-postgres-db-1 Stopped
[+] up 5/5
 ✔ Container bab-6-fluent-bit-1       Running
 ✔ Container bab-6-generator-api-1    Started
 ✔ Container bab-6-generator-auth-1   Started
 ✔ Container bab-6-generator-worker-1 Started
 ✔ Container bab-6-postgres-db-1      Healthy
[INFO] PostgreSQL berhenti; generator berjalan selama 25 detik.
[PASS] Jumlah sesudah pemulihan: 120
[INFO] Kenaikan record: 60
```

Evaluasi dan Latihan Mandiri

1. Apa risiko kehilangan log ketika Fluent Bit tidak tersedia?

   Log bisa hilang karena tidak ada collector yang menerimanya. Risiko ini dapat dikurangi dengan menyediakan buffer yang memadai pada sisi pengirim.

2. Mengapa structured logging lebih mudah dianalisis dibanding plain text logging?

   Karena data disusun dalam format terstruktur, seperti JSON. Kita bisa mencari jenis error, waktu, dan nama layanan tanpa membaca seluruh teks secara manual.

3. Kapan PostgreSQL tidak lagi cocok sebagai log storage utama?

   Ketika jumlah log sangat besar sehingga penyimpanan, pencarian, dan pemeliharaan database menjadi berat.

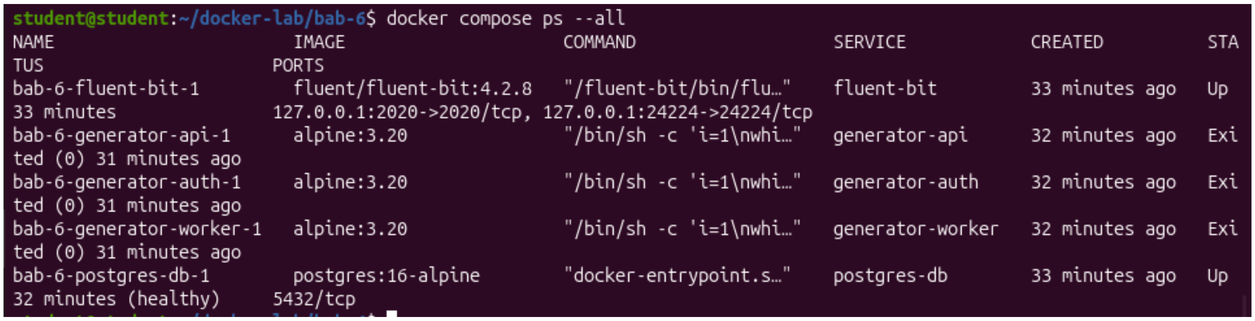
4. Apa perbedaan log untuk observability dan log untuk forensik?

   Log observability digunakan untuk memantau kondisi dan masalah sistem. Log forensik digunakan untuk menyelidiki kejadian, terutama insiden keamanan, sehingga integritas dan riwayat log sangat penting.

5. Buat query SQL untuk menampilkan jumlah log per menit dalam 10 menit terakhir.

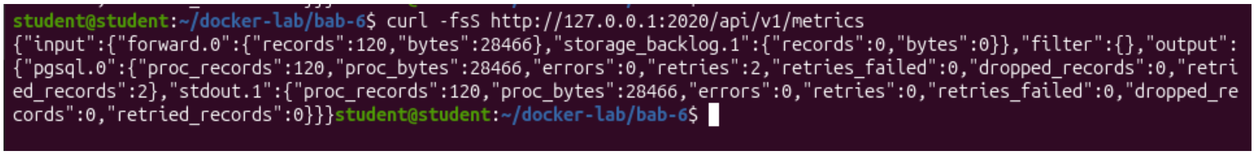
Bukti minimum:

- Screenshot docker compose ps atau docker ps yang menunjukkan service berjalan.



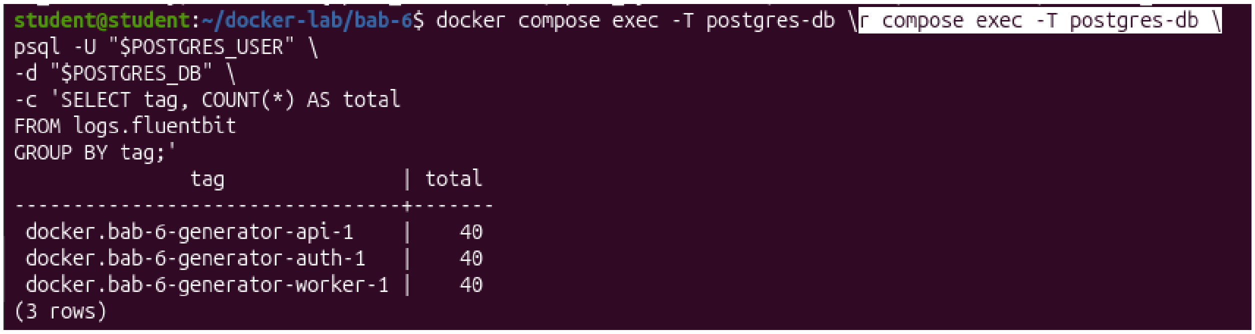
```
student@student:~/docker-lab/bab-6$ docker compose ps --all
NAME                      IMAGE                     COMMAND                    SERVICE            CREATED          STA
TUS                      PORTS
bab-6-fluent-bit-1          fluent/fluent-bit:4.2.8   "/fluent-bit/bin/flu…"   fluent-bit         33 minutes ago   Up
33 minutes               127.0.0.1:2020->2020/tcp, 127.0.0.1:24224->24224/tcp
bab-6-generator-api-1       alpine:3.20               "/bin/sh -c 'i=1\nwhi…"   generator-api      32 minutes ago   Exi
ted (0) 31 minutes ago
bab-6-generator-auth-1      alpine:3.20               "/bin/sh -c 'i=1\nwhi…"   generator-auth     32 minutes ago   Exi
ted (0) 31 minutes ago
bab-6-generator-worker-1    alpine:3.20               "/bin/sh -c 'i=1\nwhi…"   generator-worker   32 minutes ago   Exi
ted (0) 31 minutes ago
bab-6-postgres-db-1         postgres:16-alpine        "docker-entrypoint.s…"   postgres-db        33 minutes ago   Up
32 minutes (healthy)     5432/tcp
```

- Screenshot hasil curl/browser/API/dashboard sesuai target bab.



```
student@student:~/docker-lab/bab-6$ curl -fsS http://127.0.0.1:2020/api/v1/metrics
{"input":{"forward.0":{"records":120,"bytes":28466},"storage_backlog.1":{"records":0,"bytes":0}},"filter":{},"output":
{"pgsql.0":{"proc_records":120,"proc_bytes":28466,"errors":0,"retries":2,"retries_failed":0,"dropped_records":0,"retri
ed_records":2},"stdout.1":{"proc_records":120,"proc_bytes":28466,"errors":0,"retries":0,"retries_failed":0,"dropped_re
cords":0,"retried_records":0}}}student@student:~/docker-lab/bab-6$ 
```

- Cuplikan log atau query yang membuktikan sistem bekerja.



```
student@student:~/docker-lab/bab-6$ docker compose exec -T postgres-db \r compose exec -T postgres-db \
psql -U "$POSTGRES_USER" \
-d "$POSTGRES_DB" \
-c 'SELECT tag, COUNT(*) AS total
FROM logs.fluentbit
GROUP BY tag;'
               tag              | total
--------------------------------+-------
 docker.bab-6-generator-api-1    |    40
 docker.bab-6-generator-auth-1   |    40
 docker.bab-6-generator-worker-1 |    40
(3 rows)
```

Analisis wajib:

- Jelaskan satu masalah yang muncul dan cara Anda mendiagnosisnya.

  Pada saat pengujian buffering, PostgreSQL sengaja dihentikan sementara untuk menguji proses penyimpanan log. Fluent Bit mengalami kegagalan koneksi ke database dan menampilkan pesan `server closed the connection unexpectedly` serta `failed to flush chunk`.

  Untuk mendiagnosis masalah tersebut, saya memeriksa log Fluent Bit menggunakan perintah `docker compose logs --tail 200 fluent-bit`. Setelah PostgreSQL dijalankan kembali, Fluent Bit berhasil mengirim ulang log yang tertunda. Hal ini dibuktikan dengan jumlah log yang meningkat dari 60 menjadi 120 data.

- Jelaskan risiko keamanan atau operasional yang relevan pada bab ini.

  Risiko keamanan pada sistem logging adalah kemungkinan tercatatnya informasi sensitif, seperti password, token, atau data pengguna. Selain itu, jika PostgreSQL tidak tersedia dalam waktu lama, buffer Fluent Bit dapat penuh sehingga menyebabkan kehilangan log. Jika collector Fluent Bit mati dan pengirim log tidak memiliki buffering yang memadai, sebagian log juga berisiko hilang.

- Berikan rekomendasi perbaikan bila lab ini akan dibawa ke production-like environment.

  Jika sistem digunakan pada lingkungan production-like, perlu diterapkan pembatasan akses database, perlindungan informasi sensitif, backup data, serta pemantauan kapasitas penyimpanan. Selain itu, kapasitas buffer dan mekanisme retry perlu dikonfigurasi dengan baik agar log tidak mudah hilang ketika terjadi gangguan. Sistem juga perlu dilengkapi monitoring kesehatan service dan kebijakan retensi log.